

# Microbes & cells — overview

Immune cells, pathogens and body cells side by side

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The microscope-style images are not real photographs — they are stylised drawings that hint at typical shape and staining. The infographic and kids styles are likewise hand-drawn SVGs.

## Helper T cell (CD4)

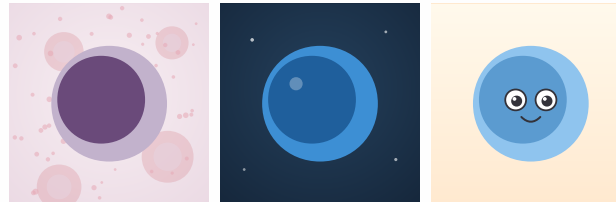


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Conductor of the immune response: recognises antigen fragments shown by other cells and releases messengers that switch other cells on.

**DEPENDS ON / TARGETS:** Needs antigen-presenting cells (dendritic cells, macrophages). Activates B cells and CD8 killer cells.

## Cytotoxic T cell (CD8)

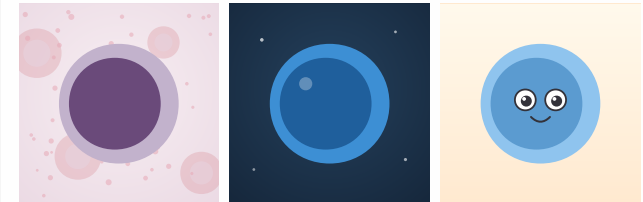


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Killer cell: spots infected or tumour cells by their MHC-I signal and kills them precisely with perforin and granzymes.

**DEPENDS ON / TARGETS:** Primed by dendritic cells, often with help from CD4 T cells. Targets virus-infected body cells and tumour cells.

## B cell



MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Antibody factory: binds antigen with its B-cell receptor and matures into plasma cells that pump out tailored antibodies.

**DEPENDS ON / TARGETS:** Usually needs help from CD4 T cells to switch into high gear. Targets free bacteria, virus particles and toxins in blood and tissue.

## Plasma cell

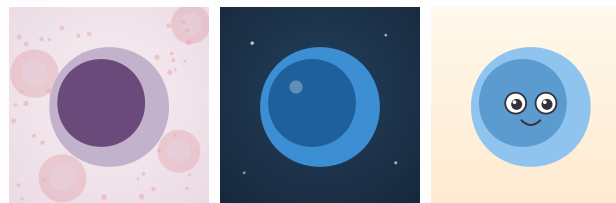


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** End stage of a B cell: a walking assembly line that secretes antibodies by the bucketload.

**DEPENDS ON / TARGETS:** Differentiates from an activated B cell. Its antibodies tag pathogens for other immune cells to deal with.

## Natural killer cell (NK)

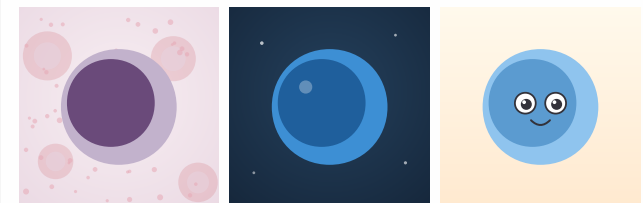


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Innate-immune bouncer: kills cells that have lost their MHC-I 'ID badge' — typical for virus infection or cancer.

**DEPENDS ON / TARGETS:** Revved up by interferons and IL-12. Kills virus-infected and tumour cells without prior priming.

## Regulatory T cell (Treg)

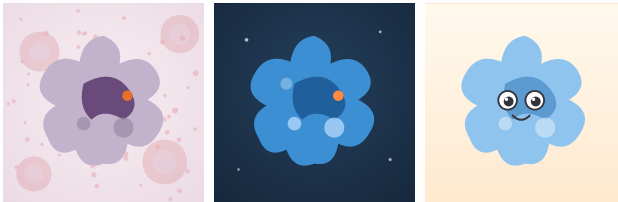


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Brake pedal of the immune system: damps other immune cells so the response doesn't overshoot and turn into autoimmunity.

**DEPENDS ON / TARGETS:** Acts on helper T cells, CD8 killers and B cells — slows them down rather than killing them.

## Macrophage

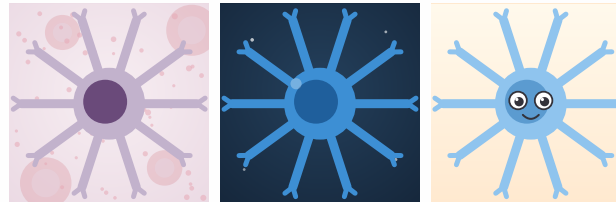


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** 'Big eater': engulfs bacteria, dead cells and debris, presents fragments to T cells and calls reinforcements with cytokines.

**DEPENDS ON / TARGETS:** Differentiates from blood monocytes. Phagocytoses bacteria, fungi, cancer cells and cleans up wounds.

## Dendritic cell

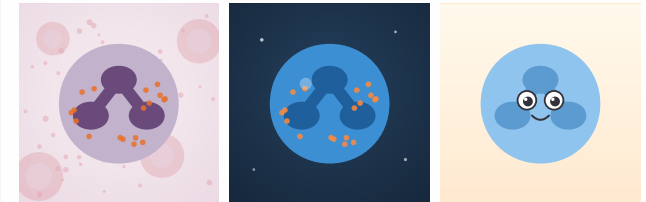


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Scout and storyteller: gathers antigen in tissues, travels to lymph nodes and shows it to T cells — kicks off the adaptive response.

**DEPENDS ON / TARGETS:** Bridges innate and adaptive immunity. Activates helper T cells and CD8 killer cells.

## Neutrophil

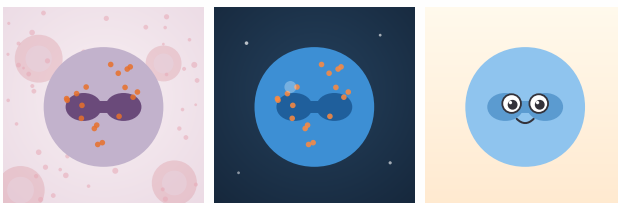


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** First responder: arrives first at sites of inflammation, eats bacteria, dumps antimicrobials and even casts sticky DNA nets (NETs).

**DEPENDS ON / TARGETS:** Chemokine-guided. Targets bacteria and fungi; lives only hours to a few days.

## Eosinophil

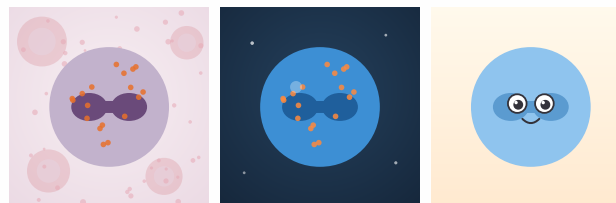


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Specialist against parasitic worms and a player in allergic reactions. Packs big red granules full of toxic proteins.

**DEPENDS ON / TARGETS:** Recruited by IL-5 from Th2 and ILC2 cells. Targets parasites and amplifies asthma and allergic reactions.

## Basophil

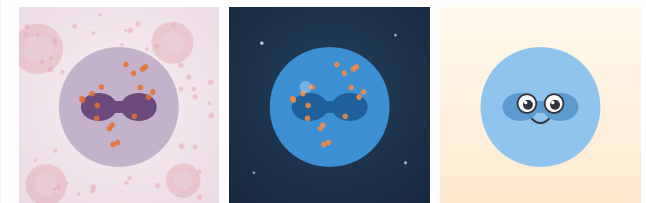


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Rare in blood but loud: releases histamine and heparin and drives allergic and inflammatory reactions.

**DEPENDS ON / TARGETS:** Responds to IgE antibodies. Involved in allergies and in defence against parasites.

## Mast cell



MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Tissue-resident sentinel packed with granules: on alarm it dumps histamine in an instant — swelling, itching, help signals.

**DEPENDS ON / TARGETS:** Triggered by IgE antibodies and pathogen signals. Drives allergy and anaphylaxis reactions.

## Cocci (round bacteria)

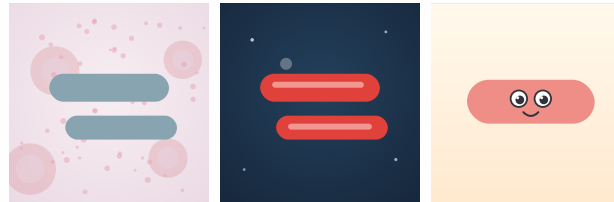


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Round bacteria, often in clusters or chains (Staphylococci, Streptococci). Some are harmless guests, others cause pneumonia, skin or throat infections.

**DEPENDS ON / TARGETS:** Phagocytosed by neutrophils and macrophages; antibodies help tag them.

## Rod bacteria

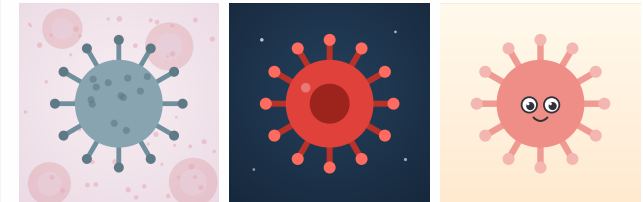


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Elongated bacteria like E. coli, Salmonella, tuberculosis bacilli. Multiply fast and can release toxins.

**DEPENDS ON / TARGETS:** Phagocytosed; intracellular species (TB) need T-cell help to be cleared.

## Virus



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**FUNCTION:** Not really alive: just genetic material in a shell. Sneaks into body cells and forces them to build new virus particles.

**DEPENDS ON / TARGETS:** Hunted by CD8 T cells and NK cells; antibodies bind free virus particles.

## Fungus

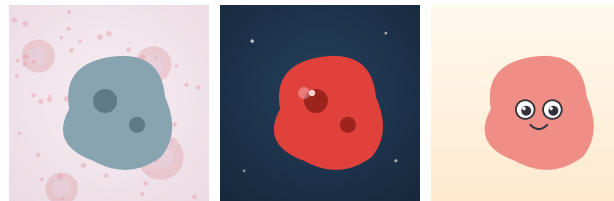


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Eukaryotic microbes — yeasts or filaments (hyphae). Examples: Candida in mouth/gut, Aspergillus in lungs.

**DEPENDS ON / TARGETS:** Controlled by neutrophils, macrophages and Th17 cells.

## Parasite

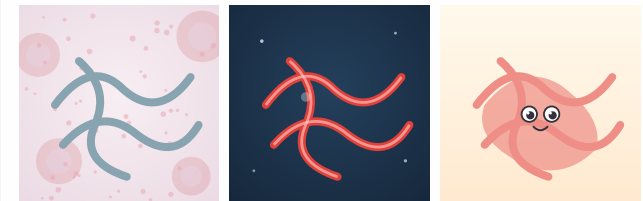


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** From single-celled (Plasmodium → malaria) to worms. Lives in or on its host and damages it in the process.

**DEPENDS ON / TARGETS:** Often fought with IgE, eosinophils and mast cells; antibodies and Th2 responses are central.

## Prion

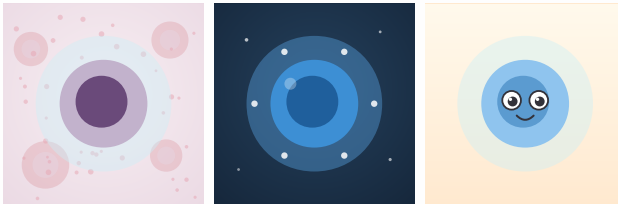


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Not alive, not a virus — a misfolded protein that forces other proteins to misfold the same way. Causes diseases like Creutzfeldt-Jakob, BSE.

**DEPENDS ON / TARGETS:** Largely invisible to the immune system; no antibody defence.

## Hematopoietic stem cell (HSC)

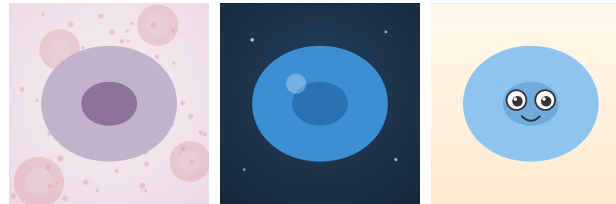


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Mother of all blood cells. Lives in bone marrow, divides rarely, self-renews and gives rise to every blood lineage.

**DEPENDS ON / TARGETS:** Needs niche cells in bone marrow (osteoblasts, stromal cells) and growth factors.

## Erythrocyte (red blood cell)



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**FUNCTION:** Oxygen carrier: no nucleus, packed with haemoglobin. Lives about 120 days and circulates millions of times.

**DEPENDS ON / TARGETS:** Production driven by the hormone erythropoietin from the kidney. Delivers oxygen to every body cell.

## Megakaryocyte

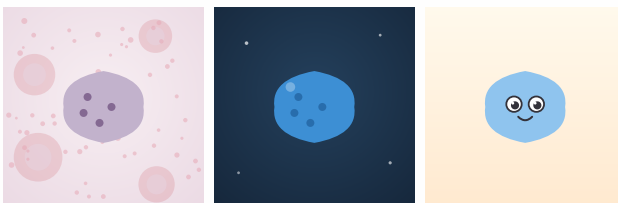


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Giant bone-marrow cell with a multi-lobed nucleus. Pinches off thousands of platelets from its cytoplasm.

**DEPENDS ON / TARGETS:** Driven by thrombopoietin (TPO) from the liver. Provides the source material for platelets.

## Thrombocyte (platelet)

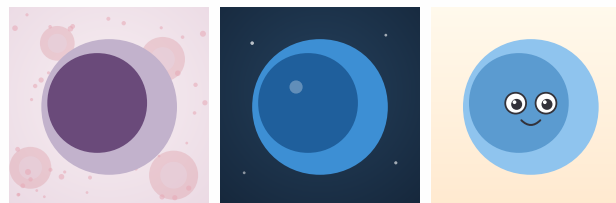


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Tiny anucleate fragments. Plug wounds, form the first clot and trigger the coagulation cascade.

**DEPENDS ON / TARGETS:** Originate from megakaryocytes. Cooperate closely with coagulation factors and endothelial cells.

## Monocyte

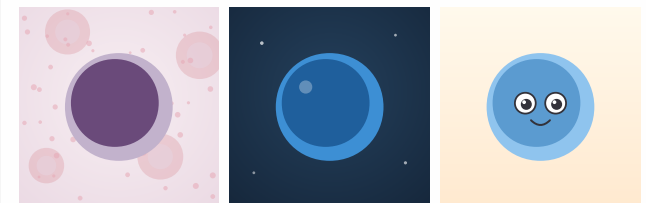


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Large blood-borne precursor. Migrates into tissues and matures there into macrophages or dendritic cells.

**DEPENDS ON / TARGETS:** Responds to inflammatory signals. Supplies tissues with fresh phagocytes.

## Lymphocyte (generic)

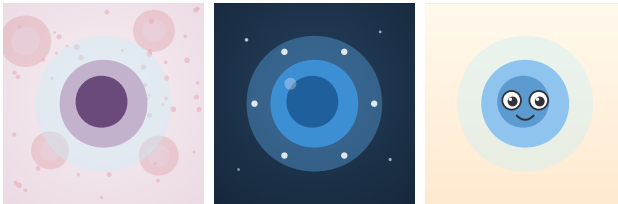


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Umbrella term for T, B and NK cells. Small, round, with a dense nucleus — the accountants of the immune system.

**DEPENDS ON / TARGETS:** Mature in bone marrow (B, NK) or thymus (T). Patrol lymph nodes, spleen and blood.

## Mesenchymal stem cell (MSC)

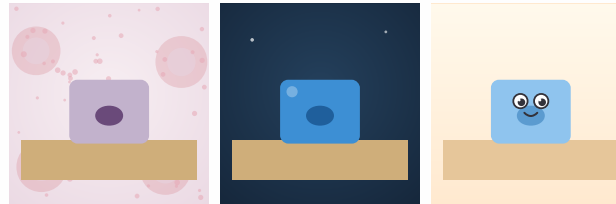


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Reserve cell for supporting tissues. Can turn into bone, cartilage, muscle, fat or connective-tissue cells.

**DEPENDS ON / TARGETS:** Lives in bone marrow, fat tissue and umbilical cord. Supplies precursors on demand.

## Osteoblast (bone builder)

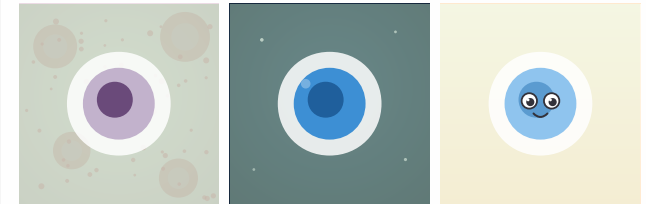


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Builds new bone: secretes collagen and mineralises it with calcium. Later gets walled in as an osteocyte.

**DEPENDS ON / TARGETS:** Balances with osteoclasts that resorb bone. Hormonally controlled by PTH and vitamin D.

## Chondrocyte (cartilage cell)

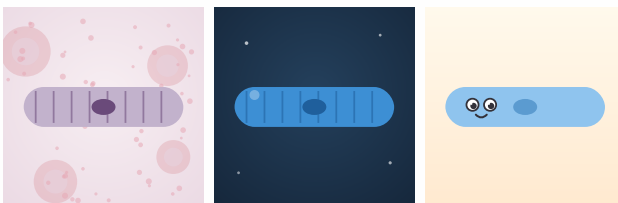


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Sits alone or in small groups inside a 'lacuna' of cartilage matrix. Produces and maintains the rubbery cartilage tissue.

**DEPENDS ON / TARGETS:** Cartilage has almost no blood supply — cells rely on diffusion. Crucial for joints, intervertebral discs, growth plates.

## Myocyte (muscle cell)

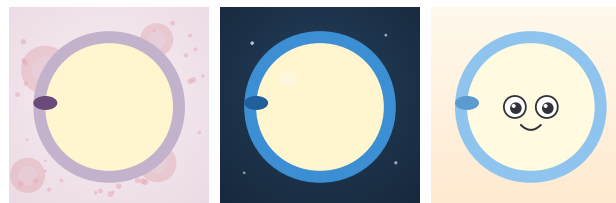


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Long, fibrous cell packed with actin/myosin filaments. Shortens on an electrical signal and generates force and movement.

**DEPENDS ON / TARGETS:** Commanded by motor neurons, fuelled by glucose and oxygen from the blood.

## Adipocyte (fat cell)

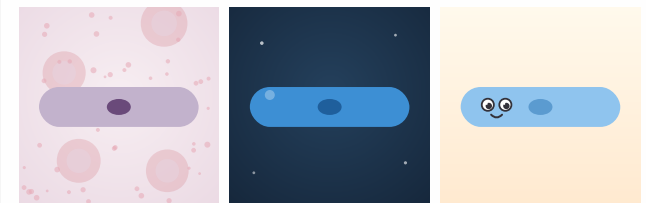


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Lipid storage: a single huge fat droplet pushes the nucleus and cytoplasm to the rim. Also acts as an endocrine cell (e.g. leptin).

**DEPENDS ON / TARGETS:** Stimulated by insulin to take up fat. Releases energy for other cells when food is scarce.

## Fibroblast

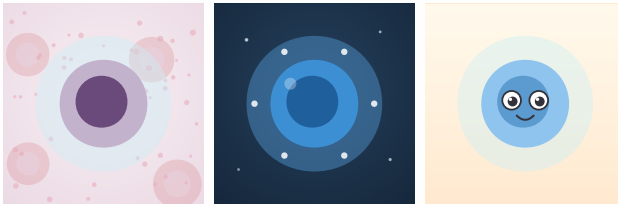


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Connective-tissue all-rounder: makes collagen and other fibres, keeps skin and organs in shape and closes wounds.

**DEPENDS ON / TARGETS:** Activated at wounds by growth factors (TGF- $\beta$ , PDGF). Works closely with macrophages.

## Neural stem cell (NSC)

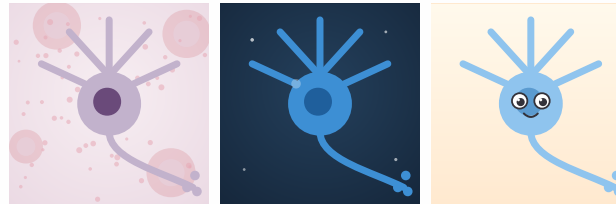


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Precursor of all neural cell types. In the adult brain only active in a few niches (e.g. hippocampus).

**DEPENDS ON / TARGETS:** Responds to growth factors (EGF, FGF) and to tissue cues.

## Neuron



MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Compute-and-signal cell: collects inputs through dendrites, sums them up and fires electrical impulses down its axon.

**DEPENDS ON / TARGETS:** Depends on astrocytes (support) and oligodendrocytes (myelin). Communicates with other neurons through synapses.

## Astrocyte



MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Star-shaped support cell. Feeds neurons, balances ions and neurotransmitters, helps form parts of the blood-brain barrier.

**DEPENDS ON / TARGETS:** Supports neurons and regulates local blood flow.

## Oligodendrocyte

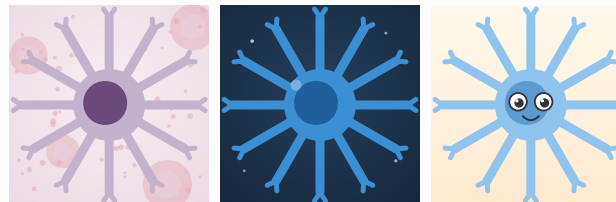


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Wraps several axons at once with myelin in brain and spinal cord — the electrical insulation of nerves.

**DEPENDS ON / TARGETS:** Partly fuels axons. In multiple sclerosis this myelin layer is the target.

## Microglia

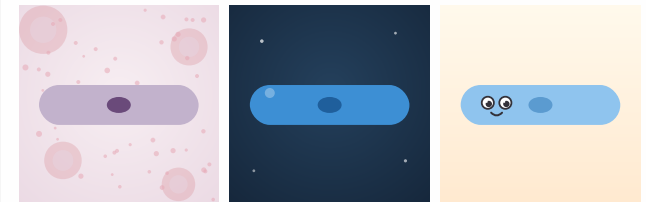


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Brain-resident macrophages: constantly probe the tissue with thin processes, clear debris, prune synapses.

**DEPENDS ON / TARGETS:** Yolk-sac derived in the embryo. First line of defence against brain infections.

## Schwann cell



MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Myelin maker in the peripheral nervous system: each Schwann cell wraps one segment of a single axon.

**DEPENDS ON / TARGETS:** Also helps peripheral nerves regenerate after injury.

## Keratinocyte (skin cell)



MICROSCOPE STYLE

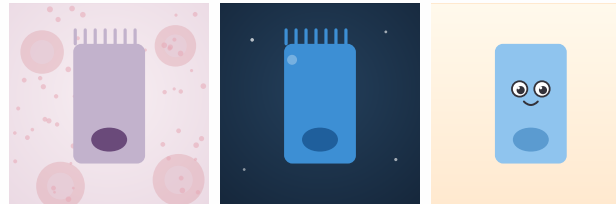
INFOGRAPHIC STYLE

KIDS STYLE

**FUNCTION:** Building block of the outer skin. Lays down keratin, cornifies upward and forms a waterproof, robust barrier.

**DEPENDS ON / TARGETS:** Renewed from stem cells in the basal layer. Cooperates with melanocytes and Langerhans cells.

## Enterocyte (gut cell)



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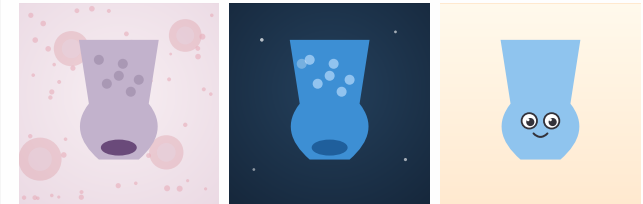
INFOGRAPHIC STYLE

KIDS STYLE

**FUNCTION:** Tall column-shaped cell with a dense brush border (microvilli). Absorbs nutrients from the gut and passes them on.

**DEPENDS ON / TARGETS:** Replaced every few days by intestinal stem cells. Forms a barrier against gut microbes.

## Goblet cell



MICROSCOPE STYLE

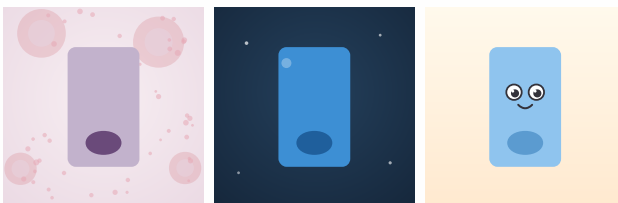
INFOGRAPHIC STYLE

KIDS STYLE

**FUNCTION:** Goblet-shaped mucus producer in gut and airways. Releases mucin packets that form a protective gel over the epithelium.

**DEPENDS ON / TARGETS:** Cooperates with enterocytes. Protects against dryness and bacterial contact.

## Paneth cell



MICROSCOPE STYLE

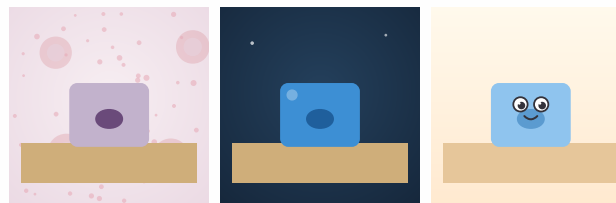
INFOGRAPHIC STYLE

KIDS STYLE

**FUNCTION:** Sits at the base of intestinal crypts and fires antimicrobial proteins (defensins, lysozyme). Protects the intestinal stem cells.

**DEPENDS ON / TARGETS:** Together with stem cells forms the crypt-base niche.

## Alveolar cell type II



MICROSCOPE STYLE

INFOGRAPHIC STYLE

KIDS STYLE

**FUNCTION:** Small cuboidal lung cell. Makes surfactant — the film that keeps alveoli open. Also acts as stem cell for type I cells.

**DEPENDS ON / TARGETS:** Lives among the thin type-I cells that handle gas exchange.

## Urothelial cell



MICROSCOPE STYLE

INFOGRAPHIC STYLE

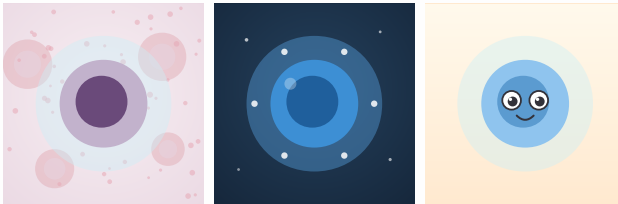
KIDS STYLE

**FUNCTION:** Stretchable cell of the bladder wall. Reshapes its surface as the bladder fills and seals against acidic urine.

**DEPENDS ON / TARGETS:** Forms a barrier between urine and body. Renewed from basal cells.



## Endothelial progenitor (EPC)

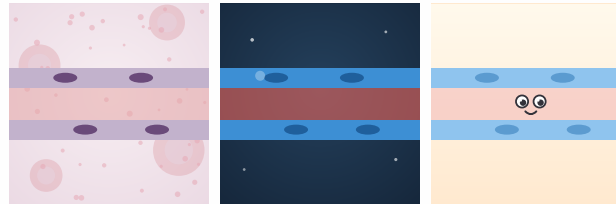


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Young endothelial cell. Mobilised during wound healing and angiogenesis to replace damaged vessel wall.

**DEPENDS ON / TARGETS:** Recruited from bone marrow; responds to VEGF.

## Endothelial cell (continuous)

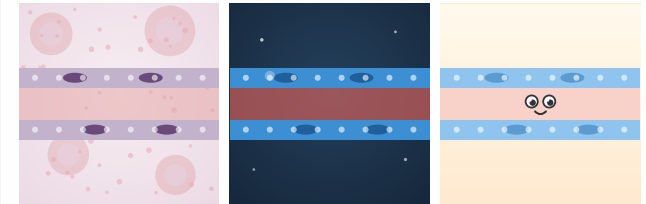


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Flat cobblestone that lines most vessels. Regulates what diffuses between blood and tissue.

**DEPENDS ON / TARGETS:** Responds to blood-flow shear and hormones (e.g. histamine).

## Fenestrated endothelium

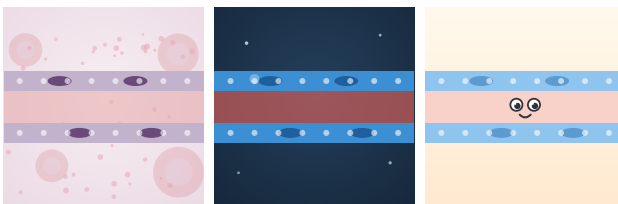


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Endothelium with little 'windows' — permeable to fluid and molecules. Found in kidney, gut and endocrine glands.

**DEPENDS ON / TARGETS:** Filters blood, releases hormones quickly or takes up nutrients.

## Sinusoidal endothelium (liver)



MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Very gappy — even large molecules pass through. Allows the liver direct contact with bloodflow.

**DEPENDS ON / TARGETS:** Works closely with hepatocytes and Kupffer cells (liver macrophages).

## Lymphatic endothelium



MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Wall of lymphatic vessels. The cells overlap like flaps so lymph flows only one way.

**DEPENDS ON / TARGETS:** Collects excess tissue fluid and returns it to the bloodstream.

## Blood-brain-barrier endothelium

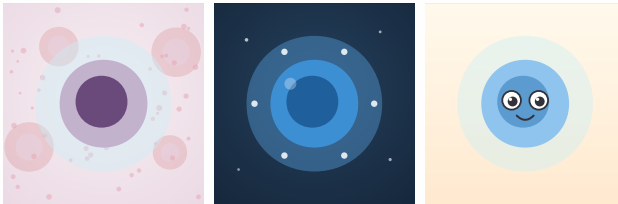


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Extra-tight: cells are locked together by tight junctions. Keeps almost everything out of the brain.

**DEPENDS ON / TARGETS:** Cooperates with astrocytes and pericytes that help form the barrier.

## iPS cell

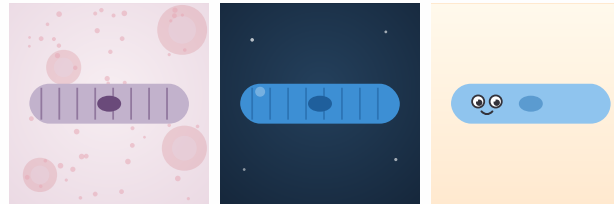


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** A body cell (e.g. skin) reset by a handful of transcription factors into an embryonic-like state — can develop into almost any cell type.

**DEPENDS ON / TARGETS:** Differentiated in vitro into a target lineage with specific growth factors.

## Cardiomyocyte (heart muscle cell)



MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Striated muscle cell that beats rhythmically. Connected by intercalated discs so the heartbeat stays in sync.

**DEPENDS ON / TARGETS:** Responds to signals from the sinoatrial node and to hormones (adrenaline).

## Hepatocyte (liver cell)

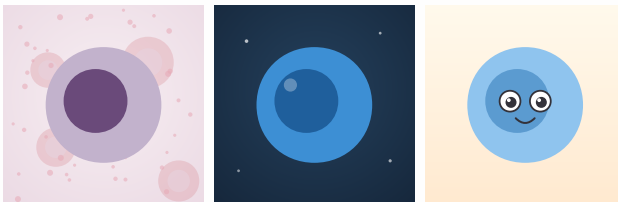


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Metabolic all-rounder: breaks down toxins, makes bile and proteins, stores glucose as glycogen.

**DEPENDS ON / TARGETS:** Receives blood from portal vein (gut) and hepatic artery. Can divide and regenerate after damage.

## Beta cell (pancreas)

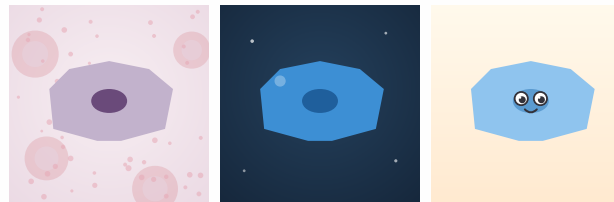


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Insulin maker in the islets of Langerhans. Senses blood glucose and releases insulin when it rises.

**DEPENDS ON / TARGETS:** Destroyed by the immune system in type-1 diabetes — replacing it from iPS cells is an active research area.

## Retinal pigment epithelium (RPE)

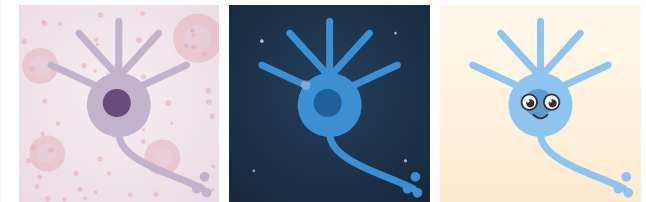


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Dark cell layer behind the photoreceptors. Recycles spent receptor tips and supplies the photoreceptors with nutrients.

**DEPENDS ON / TARGETS:** Fails in age-related macular degeneration — iPS-derived RPE cells are being tested as transplants.

## Dopaminergic neuron



MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

**FUNCTION:** Neuron that releases dopamine. Crucial for reward, motivation and movement planning.

**DEPENDS ON / TARGETS:** Dies off in Parkinson's disease — iPS-derived dopamine neurons are being explored as cell-replacement therapy.